A Controlled Comparison of a Compact CNN and ResNet18 Transfer Learning for Six-Class SAR Aircraft Classification

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Independent project repository

Abstract

We present a controlled comparison between a compact convolutional neural network trained from scratch and a ResNet18 model fine-tuned from ImageNet weights for six-class aircraft classification in synthetic aperture radar (SAR) imagery. Using the public SAR-ACD dataset (3,032 images, reduced to 2,990 after a duplicate audit), we hold every non-model factor constant: identical deduplication, an identical stratified split (2,098/446/446), identical input resolution, normalisation, augmentation, optimisation schedule, and the same three random seeds. Across three seeds the compact CNN reaches a macro F1 of 0.6233 ± 0.0291 while ResNet18 reaches 0.9147 ± 0.0046, a gain of +0.2913 (significant against the seed-noise threshold of 0.0291). Beyond accuracy, transfer learning also reduces seed-to-seed variance by roughly 6.3×. An error analysis shows that a single class pair (A220 ↔ A320321) accounts for 52.8% of all remaining mistakes, and that the two classes are statistically almost indistinguishable in size and shape. Finally, we quantify what the model attends to with Grad-CAM, and we deliberately report this evidence as indicative rather than conclusive, because the dataset provides no bounding boxes and the aircraft occupies about three quarters of each chip, which limits the discriminative power of any localisation metric. All numbers in this report are generated programmatically from the experiment artefacts.

Keywords

SAR target classification, transfer learning, ResNet18, Grad-CAM, reproducibility, error analysis

1 Introduction

Synthetic aperture radar (SAR) provides day-and-night, weather-independent imaging and is widely used for remote sensing. Recognising aircraft types in SAR imagery is a challenging fine-grained problem: the targets are small, the imagery is single-channel and dominated by strong scatterers, and the appearance of a given aircraft changes drastically with the observation azimuth [2].

This report studies a deliberately simple question: *for a small SAR aircraft dataset, how much does ImageNet transfer learning actually buy compared with a compact network trained from scratch, and where does the difference come from?* We answer it with a controlled experiment rather than a leaderboard entry. Our contributions are:

- A **controlled comparison** in which every factor except the model is held fixed, including the random seeds and the data split, so that observed differences can be attributed to the model.
- A **duplicate audit** performed *before* splitting, which removes 42 images including 20 cross-class duplicate groups; leaving them in place would leak identical images across the train and test splits [5].

Table 1: Measured dataset statistics, re-computed locally from the 3,032 files rather than copied from the dataset description.

Class	Raw	After dedup	Removed
A220	464	460	4
A320321	510	498	12
A330	512	501	11
ARJ21	514	513	1
Boeing737	528	528	0
Boeing787	504	490	14
Total	3,032	2,990	42

- An **error analysis** that localises the residual failures to a single aircraft pair, and quantifies how similar the two classes are.
- A **critical Grad-CAM analysis**, including an explicit account of why the localisation evidence is weaker than it first appears.

2 Dataset

2.1 Source and actual statistics

We use **SAR-ACD** [10], a public six-class civil aircraft dataset released by the Aerospace Information Research Institute, Chinese Academy of Sciences. Rather than trusting the published summary, we re-counted the data locally; Table 1 reports the measured values.

The images are L (single-channel) JPEGs with variable size (35–200 px wide, mean 91.6; 32–150 px high, mean 83.0); there are 635 distinct size combinations and 0 unreadable files. A fixed input size of 96×96 was therefore chosen, close to the native median size. The class ratio is 1.148:1, mildly imbalanced, which is why we report macro F1 as the primary metric. Figure 1 summarises the dataset.

2.2 Duplicate audit and splitting

A byte-level MD5 audit found 42 exact duplicates in 20 cross-class groups and 2 within-class groups. Cross-class duplicates are images that appear under two different labels; because no ground truth can arbitrate them, *both* copies are dropped. Within-class duplicates keep one copy. Deduplication is performed **before** splitting: otherwise the same image could appear in both the training and test sets, silently inflating the test score [5].

The remaining 2,990 images are divided into 2,098 training, 446 validation and 446 test images, using *stratified* sampling with a fixed seed (42). The split is written to disk and reused by every model and every seed, so that all runs are compared on identical data. After splitting, the pipeline asserts that the path sets and the content hashes of the three subsets are pairwise disjoint, and that the class proportions deviate by less than one percentage point.

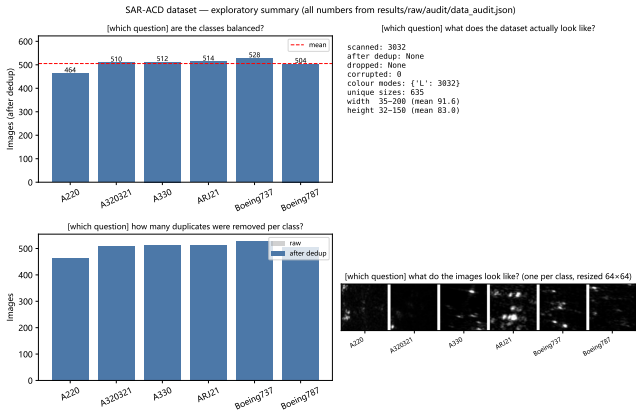


Figure 1: Dataset overview. All values are read from the audit artefact produced by the data pipeline; none are transcribed by hand.

3 Method

3.1 Compact CNN baseline

The baseline is a four-layer convolutional network designed for 96×96 single-channel input: four 3×3 convolutions (241,382 parameters) with batch normalisation [4], 2×2 max-pooling after each of the first three, global average pooling, dropout, and a linear classifier. Because the data are grayscale, the first convolution takes **one** input channel; no channel replication is used for this model.

3.2 ResNet18 transfer learning

The second model is ResNet18 [3] with weights pre-trained on ImageNet-1k [1], as distributed by `torchvision` [8]. The final fully connected layer is replaced by a Linear($512 \to 6$) layer initialised with a small standard deviation ($\sigma = 0.01$) so that its initially random outputs cannot overwhelm the pre-trained features.

Freezing policy. We freeze the shallow stages (`conv1`, `bn1`, `layer1`, `layer2`) and fine-tune the deep stages (`layer3`, `layer4`) together with the new classifier. The rationale is that early layers encode generic low-level structure that transfers, whereas late layers encode task-specific semantics that must adapt; with only 2,098 training images, updating all 11,179,590 parameters would be prone to overfitting. The frozen modules are additionally kept in evaluation mode so that their batch-norm statistics do not drift.

Learning rates. We use discriminative learning rates: 10^{-3} for the new classifier and 10^{-4} for the fine-tuned backbone stages (≈10,496,518 trainable parameters, 93.89% of the model). Optimisation uses AdamW [7] with weight decay 10^{-4} and a cosine schedule [6].

Grayscale handling. The dataset yields one-channel images. We replicate the single channel to three channels inside the model so that *every* pre-trained weight is used unchanged. The alternative — rebuilding the first convolution with one input channel by averaging the pre-trained RGB filters — avoids the mismatch between a grayscale input and a colour-pretrained stem, but introduces an arbitrary averaging step. The replication variant is the more conservative choice for a transfer-learning study and is what we report.

Table 2: Experiment configuration. Every field below is identical for the two models except the model definition itself.

Item	Value
Input size	96×96, grayscale
Split (train/val/test)	2,098 / 446 / 446, seed 42
Augmentation	rotation ±30°, flip, translate 5%, scale 0.9–1.1 (training split only)
Optimiser	AdamW, weight decay 1e-4
Learning rate	CNN 1e-3; ResNet18 head 1e-3 / backbone 1e-4
Schedule	cosine annealing
Epochs / batch	40 / 64
Early stopping	validation macro F1, patience 10
Random seeds	42, 123, 2024
Model selection	best *validation* macro F1

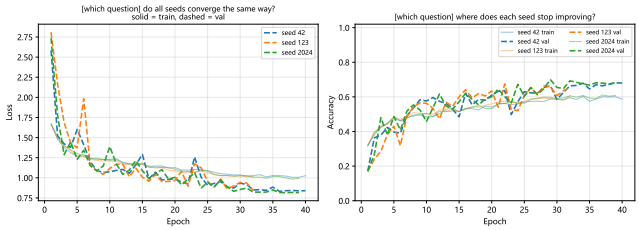


Figure 2: Compact CNN: loss and accuracy for the three seeds (solid = train, dashed = validation). The seeds diverge noticeably.

4 Experiments

4.1 Setup

All runs use the same pipeline and the same hyper-parameters (Table 2), differing only in the model. Two deliberate choices deserve emphasis. First, the normalisation statistics are computed *on the training split only* (mean 0.0572, std 0.1050) and then applied to validation and test; computing them on the full dataset would leak information. Second, we deliberately keep the *same* normalisation for both models instead of using ImageNet statistics for the pre-trained model: using ImageNet statistics is common practice, but it would change two factors at once and make the comparison unattributable.

4.2 Metrics and protocol

We report accuracy, macro F1 (primary), weighted F1, per-class precision/recall/F1, and confusion matrices (raw and row-normalised). The test split is evaluated *once* per run, using the checkpoint selected on the validation split; no test-set quantity is used for any decision. Each evaluation appends a record to a ledger, so the number of test-set evaluations is auditable. Every configuration is run with 42, 123, 2024 seeds and we report means and standard deviations.

4.3 Training behaviour

Figures 2 and 3 show the training curves of all three seeds for the two models. The baseline converges slowly and its validation score keeps fluctuating; ResNet18 reaches a much higher level within a few epochs and its three seeds nearly coincide.

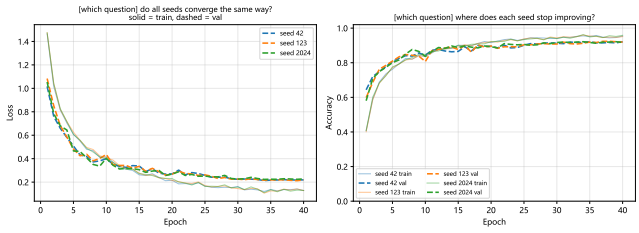


Figure 3: ResNet18 transfer learning: the three seeds track each other closely, indicating a much more stable optimisation.

Table 3: Test-set results (mean ± std over 42, 123, 2024 seeds). The seed-noise threshold is 0.0291; a difference counts as meaningful only if it exceeds it. All three differences do.

Metric	Compact CNN	ResNet18	Δ
Accuracy	0.6293 ± 0.0274	0.9163 ± 0.0042	+0.2870
Macro F1	0.6233 ± 0.0291	0.9147 ± 0.0046	+0.2913
Weighted F1	0.6250 ± 0.0295	0.9165 ± 0.0045	+0.2915

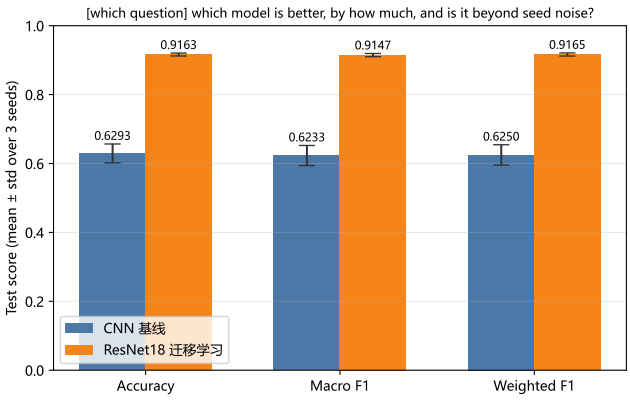


Figure 4: Overall test metrics. Error bars are standard deviations over the 42, 123, 2024 seeds; annotations give the delta and whether it exceeds the seed noise.

5 Results

5.1 Overall comparison

Table 3 reports the test-set results. ResNet18 improves macro F1 by +0.2913, from 0.6233 ± 0.0291 to 0.9147 ± 0.0046, a relative gain of 46.7%. Figure 4 shows the same comparison graphically.

Following the protocol fixed before the experiments, we only call a difference meaningful when the gap in means exceeds the larger of the two standard deviations. Here the gap (+0.2913) is about an order of magnitude larger than the threshold (0.0291), so the improvement is **significant**. A second, easily overlooked result is that transfer learning also makes the outcome far more reproducible: the macro-F1 standard deviation drops from 0.6329, 0.5839, 0.6532's spread to a spread of 0.9081, 0.9182, 0.9177 across seeds, a reduction of about 6.3×.

Table 4: Per-class F1 (mean over 42, 123, 2024 seeds), sorted by improvement.

Class	Support	CNN	ResNet18	Δ
Boeing787	73	0.5892	0.9818	+0.3926
Boeing737	79	0.6202	0.9642	+0.3440
A330	75	0.6632	0.9626	+0.2994
A220	69	0.5220	0.8213	+0.2993
ARJ21	76	0.7158	0.9662	+0.2504
A320321	74	0.6294	0.7918	+0.1624

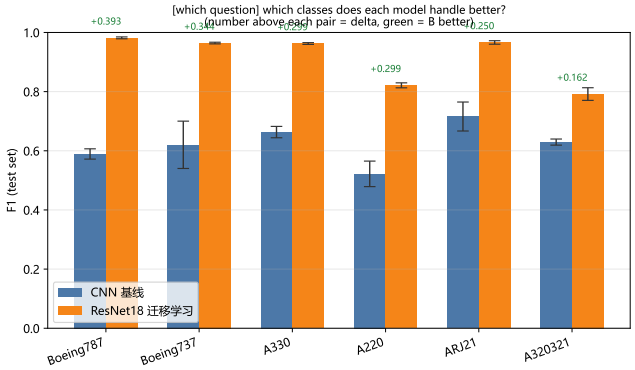


Figure 5: Per-class F1 for both models, ordered by improvement.

5.2 Per-class behaviour

Table 4 and Figure 5 break the result down by class. The improvement is not uniform. The largest gains occur precisely in the classes the baseline struggled with most, while the smallest gain is on A320321.

5.3 Confusion structure

Figure 6 contrasts the row-normalised confusion matrices. The baseline makes mistakes in many directions at once, whereas ResNet18 is close to diagonal everywhere except for one pair of classes, which we examine next.

6 Error Analysis

6.1 Final model and error sample set

The final model is chosen **on validation performance only**: we first pick the better model family by mean validation macro F1 (ResNet18 0.9199 ± 0.0011 versus the baseline 0.6802 ± 0.0118), then the best seed within that family (seed 123, validation macro F1 0.9215). No test-set quantity takes part in this selection.

On the test split the final model makes 36 errors out of 446 (8.07%). **All** of them are visualised in Figure 7; none are discarded. Errors are less confident than correct predictions (0.760 versus 0.958, a gap of 0.198), which suggests that a confidence threshold could be used for rejection in deployment.

6.2 The dominant confusion

Table 5 lists the error directions. A single pair dominates: A220 ↔ A320321 accounts for 52.8% of all errors and is *bidirectional* (0.159 of

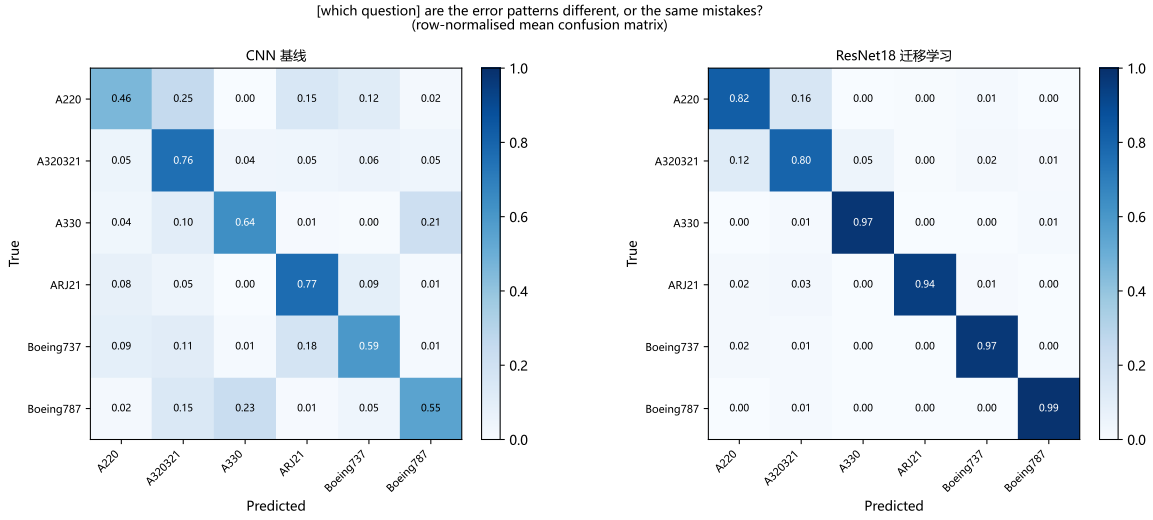


Figure 6: Row-normalised confusion matrices (mean over 42, 123, 2024 seeds). Left: compact CNN. Right: ResNet18. The baseline errs in several directions; ResNet18 leaves essentially a single confused pair.

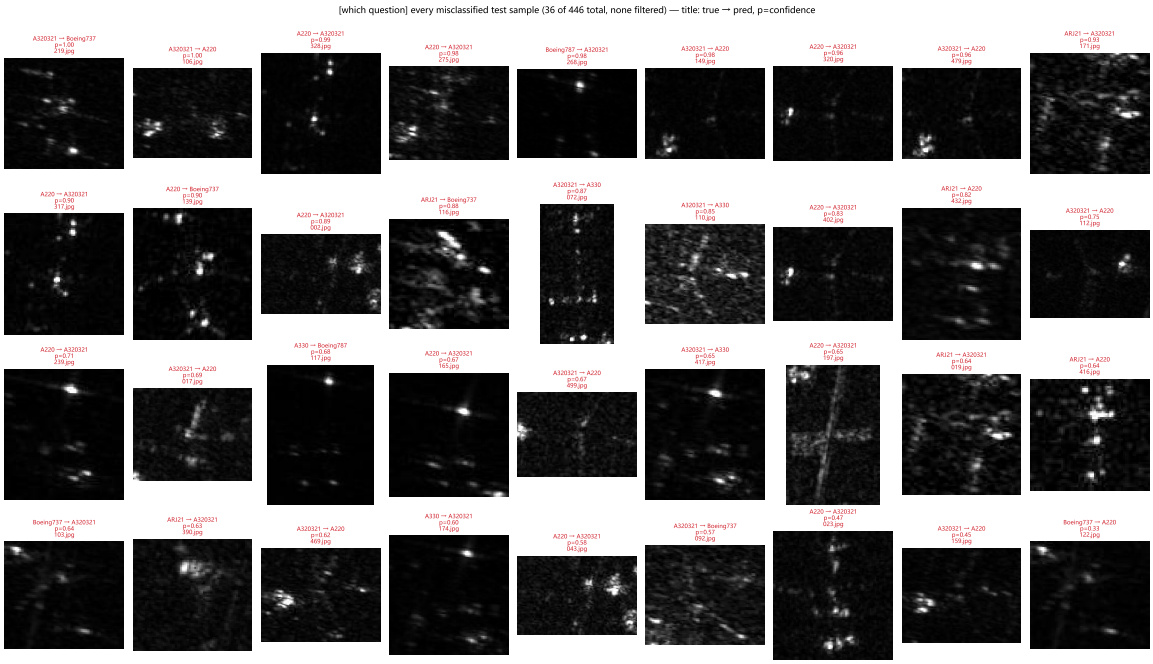


Figure 7: Every misclassified test image (36 of 446). Titles give true class → predicted class and the predicted confidence. No sample is filtered out.

true A220 predicted as A320321, and 0.108 in the opposite direction), which indicates genuine overlap in feature space rather than one class being absorbed by another.

6.3 Why are these two classes confused?

Table 6 compares measurable properties of the two classes. Every statistic differs by less than 16.6%, i.e. the two aircraft are nearly indistinguishable by size and shape statistics alone. Together with the bidirectionality of the errors, this supports three explanations, in decreasing order of evidential support:

(1) **Genuine shape similarity.** A220 and A320321 are both narrow-body airliners; at the resolution available here, their wing sweep, engine layout and fuselage proportions differ little.

Table 5: Most frequent error directions of the final model.

True → Pred.	Count	Share	Mean conf.
A220 → A320321	11	30.6%	0.785
A320321 → A220	8	22.2%	0.765
ARJ21 → A320321	3	8.3%	0.734
A320321 → A330	3	8.3%	0.789
A320321 → Boeing737	2	5.6%	0.785
ARJ21 → A220	2	5.6%	0.730

Table 6: Measured properties of the confused pair (medians unless noted). All relative differences are below 16.6%.

Statistic	A220	A320321	Rel. diff.
image width (median, px)	82	92	11.5%
image height (median, px)	72	78	8.0%
aspect ratio (median)	1.137	1.238	8.5%
bright-target area fraction	0.724	0.832	13.8%
mean image brightness	17.4	14.8	16.6%

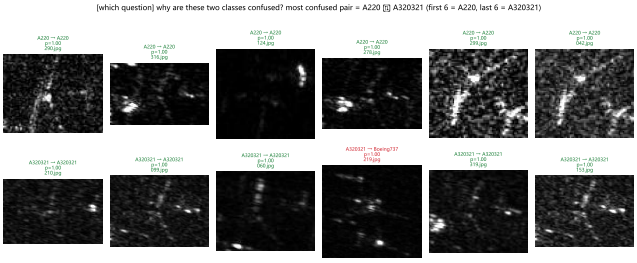


Figure 8: The most confused pair. First six images are true A220, the last six true A320321; titles give true → predicted and confidence.

(2) **Azimuth variation masquerading as class variation.** SAR scattering is extremely azimuth-sensitive: the same aircraft seen from different aspects can differ more than two different aircraft seen from the same aspect. Random rotation augmentation (±30°, applied here) may additionally blur this boundary.
(3) **Insufficient resolution.** Resizing to 96×96 discards detail; the native chips are only about 91.6×83.0 px.

The first two explanations are *inferences* consistent with the evidence, not established facts; confirming them would require azimuth metadata per image or target boxes for alignment.

7 Grad-CAM Interpretability

7.1 How the measurement is made

This dataset has no bounding boxes, so we approximate the target region by the pixels brighter than $\mu + \sigma$ (SAR aircraft are strong scatterers). We then measure the fraction of Grad-CAM [9] mass that falls on those pixels and compare it with the fraction that a random heat map would produce.

A methodological finding is important here. Our first measurement suggested an essentially random localisation. The cause was resolution, not indifference: ResNet18 downsamples by a factor of 32, so on 96×96 inputs the `layer4` feature map is only 3×3 (each cell spanning 32×32 pixels of the input). Table 7 therefore also reports `layer3` (6×6), and Figure 9 shows both.

Table 7: Grad-CAM localisation by target layer. "Lift" is the ratio of CAM mass on bright pixels to the random baseline.

Layer	CAM grid	On target	Random	Lift
`layer3`	6×6	0.0878	0.0653	1.34×
`layer4`	3×3	0.0774	0.0653	1.18×

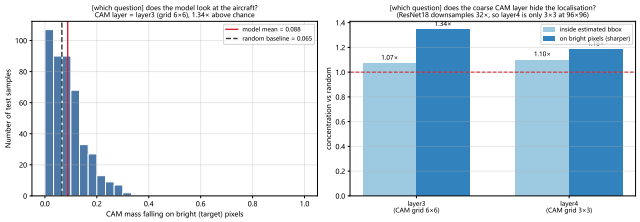


Figure 9: Left: distribution of CAM mass landing on bright target pixels (`layer3`), with the random baseline. Right: concentration relative to chance for both candidate layers, showing that the coarse `layer4` map understates the localisation.

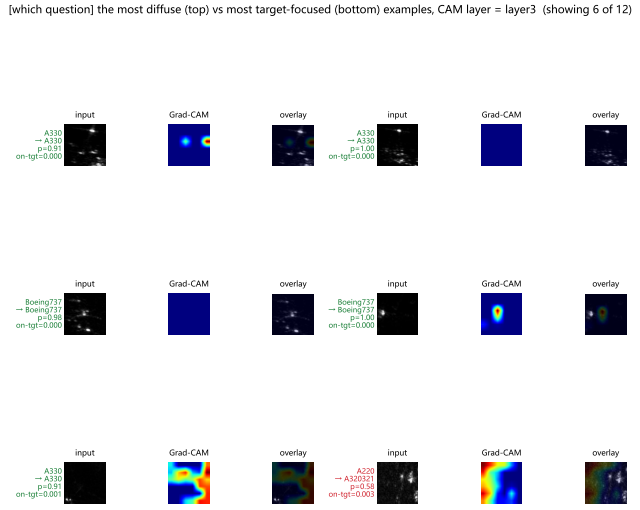


Figure 10: The most target-focused (bottom) and most diffuse (top) examples for `layer3`.

7.2 What the model attends to

Using the finer `layer3` layer, 0.0878 of the CAM mass falls on bright target pixels against a random baseline of 0.0653, a lift of **1.34×**. The same quantity is 0.0887 for correct predictions and 0.0770 for incorrect ones. Figure 9 shows the distribution and the layer comparison; Figure 10 contrasts the most target-focused and the most diffuse examples.

We deliberately stop short of claiming that the model "looks at the aircraft". A lift of 1.34× is a tendency, not a decisive result, and the measurement has three limitations: (i) the target region

is approximated by bright pixels rather than annotated, so strong sidelobes count as target; (ii) the aircraft occupies roughly three quarters of each chip, which leaves little room for a background-driven model and makes any localisation metric intrinsically weak; and (iii) even 6×6 is a coarse grid. We therefore report this section as *indicative evidence*.

8 Limitations

The public subset contains expanded samples. The released portion of SAR-ACD is described by its authors as "3,032 images in six civilian categories *after expansion*". Augmented or synthetically enlarged samples can share near-duplicate structure with the originals. Although our byte-level audit removed exact duplicates, it cannot detect near-duplicates created by augmentation, so the reported test scores should be read as an *upper bound* for genuinely unseen data.

Results do not establish cross-airport generalisation. The imagery covers the Shanghai Hongqiao airport area and a small number of other airports, acquired at particular times and geometries. Several classes may be correlated with the acquisition site or the flight schedule rather than with the aircraft shape. Our evaluation is a random split within this collection, so it measures *within-collection* generalisation only. Nothing here supports a claim about performance at a new airport, a new sensor, or a new frequency band.

Other limitations. The dataset has no bounding boxes, so localisation evidence remains indirect (Section 7) and target-aligned comparisons are impossible. The images are single-polarisation and single-channel, so polarimetric or multi-look cues are unavailable. Finally, three seeds are the minimum needed to estimate spread; the resulting standard deviations are indicative rather than precise.

9 Conclusion and Future Work

In a fully controlled comparison on six-class SAR aircraft classification, ImageNet transfer learning improved macro F1 from 0.6233 ± 0.0291 to 0.9147 ± 0.0046 (+0.2913) while reducing seed-to-seed variance by roughly 6.3×. The residual errors are highly concentrated: A220 ↔ A320321 alone accounts for 52.8% of them, and the two classes are statistically almost identical in size and shape. A critical Grad-CAM analysis indicates that the model is more target-focused than chance, but we report that as a tendency rather than a conclusion, and we explain why the measurement cannot be stronger with the data available.

Future work, in priority order:

- **Resolve the dominant confusion.** If azimuth metadata can be recovered, stratify the error rate by aspect angle; otherwise, obtain target boxes and evaluate after alignment, so that shape rather than pose is compared.
- **Near-duplicate detection.** Replace exact-hash auditing with perceptual hashing or embedding-based deduplication to bound the effect of the expanded public subset.
- **Cross-airport evaluation.** Partition by acquisition site and report leave-one-site-out accuracy, which is the claim that actually matters for deployment.
- **Freezing-policy ablation.** The implementation already supports full fine-tuning and linear probing; comparing them would separate "pre-training helps" from "more parameters help".
- **Resolution study.** Repeat at 96 and at higher input sizes to test whether the A220/A320321 boundary is limited by resolution.

Reproducibility

Every number, table and figure in this report is generated programmatically from the artefacts under `results/`. The split is fixed by seed 42 and asserted to be leakage-free; each test-set evaluation is recorded in a ledger; and each run stores its resolved configuration, log, per-epoch history, metrics and checkpoint. A clean-room reproduction consists of cloning the repository, creating the environment from `environment.yml`, and running the pipeline scripts in order.

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